Virtual machine (VM) migration is a fundamental process in virtualization that enables the movement of a virtual environment from one physical host to another to ensure performance, reliability, and efficiency.

1. Definition of VM Migration

VM migration is the process of moving the state of a virtual machine from one physical host server to another physical host 1, 2. This involves transferring the VM's configuration, memory state, and sometimes its disk files to a new hardware environment where it can resume operation 2.

2. Types of Migration

- **Live (Hot) Migration:**
- This occurs while the VM is **powered ON**.
- The state of the source host is copied to the target host while the VM is still running.
- Once the state is synchronized, the source VM is briefly suspended, and the entire VM moves to the target host to resume operation 2.
- **Requirement:** Typically requires **shared storage** or advanced mechanisms for synchronizing memory and storage states in real-time 2.
- **Non-live (Regular/Cold) Migration:**
- This occurs while the VM is **powered OFF**.
- The VM's configuration files, log files, and virtual disks are physically moved or copied to the target host 2.
- **Effect:** It results in **longer downtime** because the VM must be shut down and restarted, but it can be performed without the need for shared storage 2.

3. Migration Workflow

While the sources do not provide a numbered step-by-step list, the following workflow is synthesized from the descriptions of the process:

1. **Identification:** The need for migration is identified (e.g., a host is overloaded or requires maintenance) 2.
2. **State Copying:** For live migration, the active memory and execution state are copied to the target host while the VM remains active 2.
3. **Suspension:** The VM on the source host is briefly suspended to finalize the transfer of the most recent state changes 2.
4. **Transfer Completion:** The remaining data and control are moved to the target host 2.
5. **Resumption:** The VM is resumed or started on the new physical host 2.

4. Advantages and Limitations

Advantages:

- **Load Balancing:** Distributes workloads evenly across physical servers to prevent any single host from being overwhelmed 2, 3.

- **High Availability:** If a physical host shows signs of failure, VMs can be migrated to healthy hosts to avoid crashes 2, 4.
- **Maintenance & Upgrading:** Allows administrators to upgrade physical hardware or software without shutting down the applications running in the VMs 2, 5.
- **Disaster Recovery:** Facilitates moving critical workloads to different locations or hosts during a disaster 5.
- **Energy Efficiency:** Workloads can be consolidated onto fewer servers during low-demand periods, allowing idle servers to be shut down to save power 6.

Limitations/Challenges:

- **Migration Latency:** The process of moving a VM can cause temporary performance degradation or "latency" 7.
- **Complexity:** Managing migrations requires complex monitoring and management software 7, 8.
- **Hardware Requirements:** Live migration often demands expensive shared storage infrastructure 2.

5. Use Cases in Cloud Computing

- **Dynamic Resource Management:** Cloud providers like AWS use migration to ensure that if a physical server becomes overloaded during traffic spikes (e.g., Amazon's Black Friday), the load can be balanced by moving VMs to underutilized hosts 9, 10.
- **Scheduled Maintenance:** Moving customer VMs off a host that needs a hardware replacement without the customer experiencing service interruption 2.
- **Auto-scaling:** In conjunction with auto-scaling, migration helps in re-distributing new VM instances across a cluster for optimal performance 11.

Exam-Ready Explanation (10-Mark Answer)

Structure your answer as follows:

1. **Introduction:** Define VM migration as moving a VM's state between physical hosts to improve **resource utilization** and **availability** 1, 2.
2. **Types of Migration:** Detail the differences between **Live (Hot)** and **Cold (Regular)** migration, focusing on the power status (ON vs. OFF) and the requirement for shared storage in live scenarios 2.
3. **The "Why" (Needs):** Explain that migration is driven by the need for **upgrading hardware**, **load balancing**, and **avoiding VM failures** 2.
4. **Benefits:** Use a bulleted list to cover **High Availability**, **Energy Efficiency** (consolidation), and **Disaster Recovery Support** 4-6.
5. **Challenges:** Mention **Migration Latency** and the **Computational Overhead** required for continuous monitoring 7.
6. **Cloud Application:** Use the **Amazon/AWS example** to explain how migration supports massive e-commerce traffic surges by redistributing workloads dynamically 10, 12.
7. **Conclusion:** Summarize that while migration adds complexity, it is the key to achieving the **scalability and flexibility** required in modern data centers 5, 8.